

● MEDIUM

● **ADVANCED MATH**

Advanced Math

16

10 questions · ~15 min

Q1

ADVANCED MATH

$$D = 5,640(1.9)^t$$

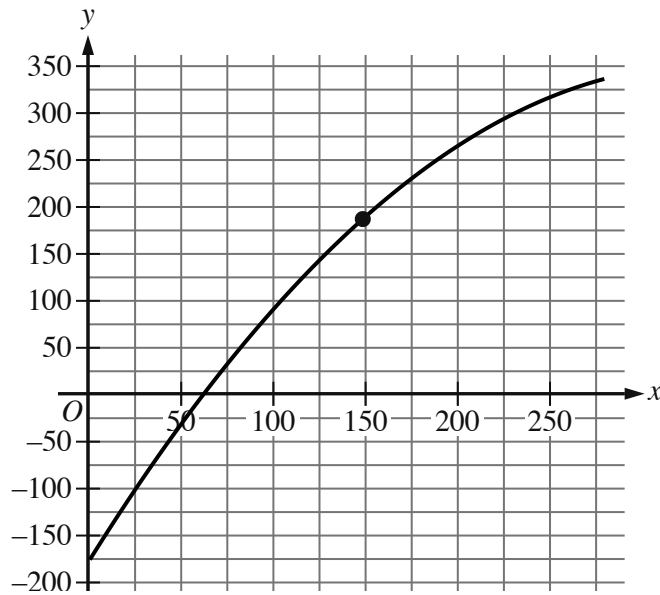
The equation above estimates the global data traffic D , in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

- A. The estimated amount of increase of data traffic, in terabytes, each year
- B. The estimated percent increase in the data traffic, in terabytes, each year
- C. The estimated data traffic, in terabytes, for the year that is t years after 2010
- D. The estimated data traffic, in terabytes, in 2010

$$N(d) = 115(0.90)^d$$

The function N defined above can be used to model the number of species of brachiopods at various ocean depths d , where d is in hundreds of meters. Which of the following does the model predict?

- A. For every increase in depth by 1 meter, the number of brachiopod species decreases by 115.
- B. For every increase in depth by 1 meter, the number of brachiopod species decreases by 10%.
- C. For every increase in depth by 100 meters, the number of brachiopod species decreases by 115.
- D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.



- The parabola opens downward.
- The parabola passes through the following approximate points:
 - (50 comma negative 33.83)
 - (149.02 comma 186.05)
 - (250 comma 315.37)

The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point 149.02186.05?

- A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
- B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

Q4

ADVANCED MATH

A rectangle has a length of x units and a width of $x - 15$ units. If the rectangle has an area of 76 square units, what is the value of x ?

- A. 4
- B. 19
- C. 23
- D. 76

$$f(t) = 5000.5^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

- A. 500
- B. 12
- C. 5
- D. 2

$$f(x) = (x + 4)(x - 1)(2x - 3)$$

The function f is defined above. Which of the following is NOT an x -intercept of the graph of the function in the xy -plane?

- A. $(-4, 0)$
- B. $\left(-\frac{2}{3}, 0\right)$
- C. $(1, 0)$
- D. $\left(\frac{3}{2}, 0\right)$

Q7

ADVANCED MATH

According to Moore's law, the number of transistors included on microprocessors doubles every 2 years. In 1985, a microprocessor was introduced that had 275,000 transistors. Based on this information, in which of the following years does Moore's law estimate the number of transistors to reach 1.1 million?

- A. 1987
- B. 1989
- C. 1991
- D. 1994

Q8

ADVANCED MATH

A company opens an account with an initial balance of \$ 36,100.00. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A , where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is \$ 68,071.93. Which equation could define A ?

- A. $A(t) = 36,100.001.05^t$
- B. $A(t) = 31,971.931.05^t$
- C. $A(t) = 31,971.930.05^t$
- D. $A(t) = 36,100.000.05^t$

Q9

ADVANCED MATH

The function p is defined by $pn = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

- A. 2
- B. $\frac{8}{3}$
- C. 7
- D. 8

Q10

ADVANCED MATH

$$gx = x^2 + 55$$

What is the minimum value of the given function?

- A. 0
- B. 55
- C. 110
- D. 3,025